

Consider the following Quadratic Optimization with an equality constraint:

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1^2 + 4x_2^2 - 3x_1 + x_2 + 3 \\ \text{s. t.} \quad & 2x_1 - 3x_2 = 4 \end{aligned}$$

Use the Lagrangian function to derive a set of linear equations the solution to which yields the optimal point x_1^*, x_2^* . (Do not need to solve the equations)

Solution:

$$L(x, v) = x_1^2 + 4x_2^2 - 3x_1 + x_2 + 3 + v(2x_1 - 3x_2 - 4)$$

$$\nabla L(x^*, v^*) = 0$$

$$2x_1^* - 3x_2^* = 4$$

$$2x_1^* + 2v^* = 3$$

$$8x_2^* - 3v^* = -1$$